

科研方向综述导读 · 中英对照版

编制：Ahsoka 日期：2026年9月22日

对象：郑博文（应化2504）及队友

说明

这是《科研方向综述导读》的中英对照版：每篇综述给出英文原标题 + 中文译名 + DOI。

只译标题——目的是让你：

1. 快速扫读：中文一眼看懂这篇讲什么
2. 精确检索：拿英文原标题去数据库/搜索引擎找原文
3. 点 DOI 直达：链接进官方页看摘要

⚠ 译文是我的翻译，非官方译名。化学术语已按惯例校准

（如 MXene、MOF、SERS、赝电容、单原子催化等），但个别表述可能与期刊中文译法不同。

检索请以英文原标题为准。

全文获取：多数需机构订阅。可先看摘要（DOI 页免费），
或用学校图书馆 VPN 下载全文。

目录

1. 电催化 / 电解水制氢（8 篇）
 2. 锂/钠/锌离子电池（9 篇）
 3. 燃料电池（9 篇）
 4. 超级电容器（9 篇）
 5. 光催化 / 光电催化（9 篇）
 6. 多相催化 / 工业催化（9 篇）
 7. 水滑石 / 层状材料（二维材料）（9 篇）
 8. 纳米材料合成（9 篇）
 9. 高分子 / 功能材料（9 篇）
 10. 分析化学 / 传感检测（7 篇）
 11. MOF / 多孔材料（9 篇）
 12. 计算化学 / 理论模拟（9 篇）
-

1. 电催化 / 电解水制氢

1. 水电解：从教科书知识到最新科学策略与工业进展

Water electrolysis: from textbook knowledge to the latest scientific strategies and industrial developments

2022 · Chemical Society Reviews · 引用 1,968 · [DOI](https://doi.org/10.1039/d0cs01079k)

2. 用于全解水与混合水分解的双功能电催化剂

Bifunctional Electrocatalysts for Overall and Hybrid Water Splitting

2024 · Chemical Reviews · 引用 841 · [DOI](https://doi.org/10.1021/acs.chemrev.3c00332)

3. 析氧/氧还原反应催化剂：从原位监测与反应机理到理性设计

Oxygen Evolution/Reduction Reaction Catalysts: From In Situ Monitoring and Reaction Mechanisms to Rational Design

2023 · Chemical Reviews · 引用 789 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00515)

4. 电催化中的高熵合金：从基础到应用

High-entropy alloys in electrocatalysis: from fundamentals to applications

2023 · Chemical Society Reviews · 引用 530 · [DOI](https://doi.org/10.1039/d3cs00557g)

5. 金属有机框架作为太阳能全解水光催化剂

Metal – Organic Frameworks as Photocatalysts for Solar-Driven Overall Water Splitting

2022 · Chemical Reviews · 引用 527 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00460)

6. 无贵金属析氢催化剂的设计与应用

Precious Metal Free Hydrogen Evolution Catalyst Design and Application

2024 · Chemical Reviews · 引用 486 · [DOI](https://doi.org/10.1021/acs.chemrev.3c00712)

7. 纳米颗粒电催化析氢的最新进展

Recent Advances in Electrocatalytic Hydrogen Evolution Using Nanoparticles

2019 · Chemical Reviews · 引用 3,146 · [DOI](https://doi.org/10.1021/acs.chemrev.9b00248)

8. 析氧电催化剂的设计基础综述

A review on fundamentals for designing oxygen evolution electrocatalysts

2020 · Chemical Society Reviews · 引用 2,905 · [DOI](https://doi.org/10.1039/c9cs00607a)

2. 锂/钠/锌离子电池

9. 钠离子电池的醚基电解质

Ether-based electrolytes for sodium ion batteries

2022 · Chemical Society Reviews · 引用 598 · [DOI](https://doi.org/10.1039/d1cs00948f)

10. 高能锂电池电解质与界面相的设计

Designing electrolytes and interphases for high-energy lithium batteries

2023 · Nature Reviews Chemistry · 引用 465 · [DOI](https://doi.org/10.1038/s41570-023-00557-z)

11. 高性能水系锌离子电池的先进电解质

Advanced electrolytes for high-performance aqueous zinc-ion batteries

2024 · Chemical Society Reviews · 引用 451 · [DOI](https://doi.org/10.1039/d4cs00584h)

12. 钠离子电池的回收

Recycling of sodium-ion batteries

2023 · Nature Reviews Materials · 引用 435 · [DOI](https://doi.org/10.1038/s41578-023-00574-w)

13. 钠离子电池高性能层状氧化物正极的实现路径

Routes to high-performance layered oxide cathodes for sodium-ion batteries

2024 · Chemical Society Reviews · 引用 419 · [DOI](https://doi.org/10.1039/d3cs00929g)

14. 宽温域钠离子电池的新兴化学

Emerging Chemistry for Wide-Temperature Sodium-Ion Batteries

2024 · Chemical Reviews · 引用 348 · [DOI](https://doi.org/10.1021/acs.chemrev.3c00728)

15. 钠离子电池的成本与资源分析

A cost and resource analysis of sodium-ion batteries

2018 · Nature Reviews Materials · 引用 2,594 · [DOI](https://doi.org/10.1038/natrevmats.2018.13)

16. 二维MXene在能量转换与存储系统中的应用

Applications of 2D MXenes in energy conversion and storage systems

2018 · Chemical Society Reviews · 引用 1,945 · [DOI](https://doi.org/10.1039/c8cs00324f)

17. 水系锌离子电池活性材料：合成、晶体结构、形貌与电化学

Active Materials for Aqueous Zinc Ion Batteries: Synthesis, Crystal Structure, Morphology, and Electrochemistry

2020 · Chemical Reviews · 引用 1,805 · [DOI](https://doi.org/10.1021/acs.chemrev.9b00628)

3. 燃料电池

18. 质子交换膜水电解的最新进展

Recent advances in proton exchange membrane water electrolysis

2023 · Chemical Society Reviews · 引用 691 · [DOI](https://doi.org/10.1039/d2cs00681b)

19. 阴离子交换膜水电解槽与燃料电池

Anion-exchange membrane water electrolyzers and fuel cells

2022 · Chemical Society Reviews · 引用 567 · [DOI](https://doi.org/10.1039/d2cs00038e)

20. 碱性介质中的电催化与碱性膜能源技术

Electrocatalysis in Alkaline Media and Alkaline Membrane-Based Energy Technologies

2022 · Chemical Reviews · 引用 538 · [DOI](https://doi.org/10.1021/acs.chemrev.1c00331)

21. 钙钛矿型ABO₃氧化物在光催化、电催化与固体氧化物燃料电池中的应用：现状与展望

Perovskite Type ABO₃ Oxides in Photocatalysis, Electrocatalysis, and Solid Oxide Fuel Cells: State of the Art and Future Prospects

2025 · Chemical Reviews · 引用 333 · [DOI](https://doi.org/10.1021/acs.chemrev.4c00553)

22. 基于丰产金属的单原子催化剂及其能源相关应用

Single Atom Catalysts Based on Earth-Abundant Metals for Energy-Related Applications

2024 · Chemical Reviews · 引用 300 · [DOI](https://doi.org/10.1021/acs.chemrev.4c00155)

23. 氧电催化中的吸附能

Adsorption Energy in Oxygen Electrocatalysis

2022 · Chemical Reviews · 引用 299 · [DOI](https://doi.org/10.1021/acs.chemrev.1c01003)

24. 理解氧还原反应中的催化活性趋势

Understanding Catalytic Activity Trends in the Oxygen Reduction Reaction

2018 · Chemical Reviews · 引用 2,775 · [DOI](https://doi.org/10.1021/acs.chemrev.7b00488)

25. 高温质子交换膜燃料电池：先进材料与关键技术进展

High temperature proton exchange membrane fuel cells: progress in advanced materials and key technologies

2020 · Chemical Society Reviews · 引用 788 · [DOI](https://doi.org/10.1039/d0cs00296h)

26. 用于电催化氧还原的碳材料缺陷

Defects on carbons for electrocatalytic oxygen reduction

2018 · Chemical Society Reviews · 引用 598 · [DOI](https://doi.org/10.1039/c7cs00690j)

4. 超级电容器

27. MXene化学、电化学与储能应用

MXene chemistry, electrochemistry and energy storage applications

2022 · Nature Reviews Chemistry · 引用 1,492 · [DOI](https://doi.org/10.1038/s41570-022-00384-8)

28. 柔性及便携式电化学储能器件的机遇：将关注点扩展至半固态/固态电解质

Opportunities of Flexible and Portable Electrochemical Devices for Energy Storage: Expanding the Spotlight onto Semi-solid/Solid Electrolytes

2022 · Chemical Reviews · 引用 337 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00196)

29. 用于高性能超级电容器的多孔有机聚合物

Porous organic polymers for high-performance supercapacitors

2022 · Chemical Society Reviews · 引用 283 · [DOI](https://doi.org/10.1039/d2cs00065b)

30. 电化学中常相位元件（CPE）的使用

On the use of a constant phase element (CPE) in electrochemistry

2022 · Current Opinion in Electrochemistry · 引用 260 · [DOI](https://doi.org/10.1016/j.coelec.2022.101133)

31. 纳米限域中离子液体的理论与模拟

Theory and Simulations of Ionic Liquids in Nanoconfinement

2023 · Chemical Reviews · 引用 152 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00728)

32. 生物炭衍生碳材料的各种环境应用：系统综述

Biochar derived carbonaceous material for various environmental applications: Systematic review

2022 · Environmental Research · 引用 117 · [DOI](https://doi.org/10.1016/j.envres.2022.113857)

33. 非对称超级电容器的设计与机理

Design and Mechanisms of Asymmetric Supercapacitors

2018 · Chemical Reviews · 引用 3,652 · [DOI](https://doi.org/10.1021/acs.chemrev.8b00252)

34. 以赝电容材料实现高能量密度与高功率密度

Achieving high energy density and high power density with pseudocapacitive materials

2019 · Nature Reviews Materials · 引用 1,972 · [DOI](https://doi.org/10.1038/s41578-019-0142-z)

35. 二维MXene在能量转换与存储系统中的应用

Applications of 2D MXenes in energy conversion and storage systems

2018 · Chemical Society Reviews · 引用 1,945 · [DOI](https://doi.org/10.1039/c8cs00324f)

5. 光催化 / 光电催化

36. 颗粒光催化剂太阳能分解水的最新进展与展望

Recent advances and perspectives for solar-driven water splitting using particulate photocatalysts

2022 · Chemical Society Reviews · 引用 863 · [DOI](https://doi.org/10.1039/d1cs01182k)

37. 光催化分解水

Photocatalytic water splitting

2023 · Nature Reviews Methods Primers · 引用 594 · [DOI](https://doi.org/10.1038/s43586-023-00226-x)

38. 非贵金属等离激元光催化剂

Non-Noble Plasmonic Metal-Based Photocatalysts

2022 · Chemical Reviews · 引用 583 · [DOI](https://doi.org/10.1021/acs.chemrev.1c00473)

39. 金属有机框架作为太阳能全解水光催化剂

Metal – Organic Frameworks as Photocatalysts for Solar-Driven Overall Water Splitting

2022 · Chemical Reviews · 引用 527 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00460)

40. 光电化学分解水的商业化之路：大面积器件与关键放大挑战综述

The route for commercial photoelectrochemical water splitting: a review of large-area devices and key upscaling challenges

2024 · Chemical Society Reviews · 引用 368 · [DOI](https://doi.org/10.1039/d1cs01069g)

41. 光照下析氧反应应用半导体聚合物

Semiconducting Polymers for Oxygen Evolution Reaction under Light Illumination

2022 · Chemical Reviews · 引用 340 · [DOI](https://doi.org/10.1021/acs.chemrev.1c00686)

42. 二维MXene在能量转换与存储系统中的应用

Applications of 2D MXenes in energy conversion and storage systems

2018 · Chemical Society Reviews · 引用 1,945 · [DOI](https://doi.org/10.1039/c8cs00324f)

43. 迈向实用化太阳能制氢——从人工光合“叶”到农场的挑战

Toward practical solar hydrogen production – an artificial photosynthetic leaf-to-farm challenge

2019 · Chemical Society Reviews · 引用 1,323 · [DOI](https://doi.org/10.1039/c8cs00699g)

44. 金属硫化物的最新进展：从可控合成到电催化、光催化与光电化学分解水及更广应用

Recent advances in metal sulfides: from controlled fabrication to electrocatalytic, photocatalytic and photoelectrochemical water splitting and beyond

2019 · Chemical Society Reviews · 引用 1,205 · [DOI](https://doi.org/10.1039/c8cs00664d)

6. 多相催化 / 工业催化

45. 催化用双金属位点：从双核金属位点到双金属纳米团簇与纳米颗粒

Bimetallic Sites for Catalysis: From Binuclear Metal Sites to Bimetallic Nanoclusters and Nanoparticles

2023 · Chemical Reviews · 引用 689 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00733)

46. 沸石中的金属位点：合成、表征与催化

Metal Sites in Zeolites: Synthesis, Characterization, and Catalysis

2022 · Chemical Reviews · 引用 619 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00315)

47. 多孔有机笼

Porous Organic Cages

2023 · Chemical Reviews · 引用 461 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00667)

48. 用手性催化剂实现的对映选择性自由基反应

Enantioselective Radical Reactions Using Chiral Catalysts

2022 · Chemical Reviews · 引用 427 · [DOI](https://doi.org/10.1021/acs.chemrev.1c00582)

49. 金属氧化物负载的原子、团簇与纳米颗粒催化中的金属—载体相互作用

Metal – support interactions in metal oxide-supported atomic, cluster, and nanoparticle catalysis

2024 · Chemical Society Reviews · 引用 394 · [DOI](https://doi.org/10.1039/d4cs00527a)

50. 单原子电催化剂中的助催化金属—载体相互作用

Co-catalytic metal – support interactions in single-atom electrocatalysts

2024 · Nature Reviews Materials · 引用 330 · [DOI](https://doi.org/10.1038/s41578-023-00633-2)

51. 水系电解质中铜上电化学CO₂还原的进展与展望

Progress and Perspectives of Electrochemical CO₂ Reduction on Copper in Aqueous Electrolyte

2019 · Chemical Reviews · 引用 5,162 · [DOI](https://doi.org/10.1021/acs.chemrev.8b00705)

52. 负载型金属催化剂上的选择性加氢：从纳米颗粒到单原子

Selective Hydrogenation over Supported Metal Catalysts: From Nanoparticles to Single Atoms

2019 · Chemical Reviews · 引用 1,627 · [DOI](https://doi.org/10.1021/acs.chemrev.9b00230)

53. 金属有机框架的催化与光催化

Catalysis and photocatalysis by metal organic frameworks

2018 · Chemical Society Reviews · 引用 1,516 · [DOI](https://doi.org/10.1039/c8cs00256h)

7. 水滑石 / 层状材料（二维材料）

54. MXene化学、电化学与储能应用

MXene chemistry, electrochemistry and energy storage applications

2022 · Nature Reviews Chemistry · 引用 1,492 · [DOI](https://doi.org/10.1038/s41570-022-00384-8)

55. 添加剂介导的MXene插层与表面修饰

Additive-mediated intercalation and surface modification of MXenes

- 2022 · Chemical Society Reviews · 引用 374 · [DOI](https://doi.org/10.1039/d0cs01487g)
56. 高连接度三维共价有机框架的探索：拓扑、结构与新兴应用
Exploring high-connectivity three-dimensional covalent organic frameworks: topologies, structures, and emerging applications
2024 · Chemical Society Reviews · 引用 189 · [DOI](https://doi.org/10.1039/d4cs00703d)
57. 石墨烯之外二维材料的电化学剥离
Electrochemical exfoliation of 2D materials beyond graphene
2024 · Chemical Society Reviews · 引用 170 · [DOI](https://doi.org/10.1039/d3cs00815k)
58. 过渡金属硫族化合物的大面积外延生长
Large-Area Epitaxial Growth of Transition Metal Dichalcogenides
2024 · Chemical Reviews · 引用 125 · [DOI](https://doi.org/10.1021/acs.chemrev.3c00851)
59. 用于挥发性有机物传感的二维材料系统综述
A systematic review on 2D materials for volatile organic compound sensing
2022 · Coordination Chemistry Reviews · 引用 72 · [DOI](https://doi.org/10.1016/j.ccr.2022.214502)
60. 储能用的二维金属碳化物与氮化物 (MXene)
2D metal carbides and nitrides (MXenes) for energy storage
2017 · Nature Reviews Materials · 引用 7,421 · [DOI](https://doi.org/10.1038/natrevmats.2016.98)
61. 二维MXene在能量转换与存储系统中的应用
Applications of 2D MXenes in energy conversion and storage systems
2018 · Chemical Society Reviews · 引用 1,945 · [DOI](https://doi.org/10.1039/c8cs00324f)
62. 面向智能与可穿戴电子的柔性固态超级电容器
Towards flexible solid-state supercapacitors for smart and wearable electronics
2018 · Chemical Society Reviews · 引用 1,731 · [DOI](https://doi.org/10.1039/c7cs00505a)
-

8. 纳米材料合成

63. 二维II – VI族半导体纳米片：从材料合成到光电子集成
2D II – VI Semiconductor Nanoplatelets: From Material Synthesis to Optoelectronic Integration
2023 · Chemical Reviews · 引用 198 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00436)
64. 强限域胶体铅卤钙钛矿量子点：从合成到应用
Strongly-confined colloidal lead-halide perovskite quantum dots: from synthesis to applications
2024 · Chemical Society Reviews · 引用 165 · [DOI](https://doi.org/10.1039/d4cs00077c)
65. 能量转移与电子转移之争：钙钛矿纳米晶—分子杂化体系中激发态相互作用的调控
Energy Versus Electron Transfer: Managing Excited-State Interactions in Perovskite Nanocrystal – Molecular Hybrids
2022 · Chemical Reviews · 引用 148 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00172)
66. 金属纳米晶的胶体合成：从非对称生长到对称性破缺
Colloidal Synthesis of Metal Nanocrystals: From Asymmetrical Growth to Symmetry Breaking

2022 · Chemical Reviews · 引用 143 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00468)

67. 氧化锌纳米晶的胶体合成方法

Colloidal Approaches to Zinc Oxide Nanocrystals

2022 · Chemical Reviews · 引用 123 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00456)

68. 胶体半导体纳米晶激光器与激光二极管

Colloidal Semiconductor Nanocrystal Lasers and Laser Diodes

2023 · Chemical Reviews · 引用 113 · [DOI](https://doi.org/10.1021/acs.chemrev.2c00865)

69. 金属卤化物钙钛矿纳米晶：合成、后合成修饰及其光学性质

Metal Halide Perovskite Nanocrystals: Synthesis, Post-Synthesis Modifications, and Their Optical Properties

2019 · Chemical Reviews · 引用 1,800 · [DOI](https://doi.org/10.1021/acs.chemrev.8b00644)

70. 纳米结构无机材料中的稀土掺杂

Rare-Earth Doping in Nanostructured Inorganic Materials

2022 · Chemical Reviews · 引用 1,191 · [DOI](https://doi.org/10.1021/acs.chemrev.1c00644)

71. 半导体纳米晶中的局域表面等离子共振

Localized Surface Plasmon Resonance in Semiconductor Nanocrystals

2018 · Chemical Reviews · 引用 1,045 · [DOI](https://doi.org/10.1021/acs.chemrev.7b00613)

9. 高分子 / 功能材料

72. 固态锂电池复合聚合物电解质中的颗粒填料：综述

Particles in composite polymer electrolyte for solid-state lithium batteries: A review

2022 · Particuology · 引用 130 · [DOI](https://doi.org/10.1016/j.partic.2021.04.002)

73. 锂离子电池聚合物电解质综合综述

A comprehensive review of polymer electrolyte for lithium-ion battery

2022 · Polymer Bulletin · 引用 79 · [DOI](https://doi.org/10.1007/s00289-021-04008-x)

74. 用于固态锂离子电池的PVDF-HFP复合聚合物电解质综述

Review on composite polymer electrolyte using PVDF-HFP for solid-state lithium-ion battery

2024 · Materials Today Chemistry · 引用 77 · [DOI](https://doi.org/10.1016/j.mtchem.2024.101926)

75. 锂金属电池中凝胶聚合物电解质与固态电解质界面关系综述

A Review of the Relationship between Gel Polymer Electrolytes and Solid Electrolyte Interfaces in Lithium Metal Batteries

2023 · Nanomaterials · 引用 27 · [DOI](https://doi.org/10.3390/nano13111789)

76. PVDF-HFP聚合物基体改性背景下的聚合物电解质材料综述

A review on polymer electrolyte materials in context to modifications in PVDF-HFP polymer host

2025 · Journal of Solid State Electrochemistry · 引用 19 · [DOI](https://doi.org/10.1007/s10008-025-06343-9)

77. 综述——适用于三维电池结构的固态与聚合物电解质材料及相关加工方法

2024 · Journal of The Electrochemical Society · 引用 5 · [DOI](https://doi.org/10.1149/1945-7111/ad318c)

78. 为安全、高能量密度电池设计固态电解质

Designing solid-state electrolytes for safe, energy-dense batteries

2020 · Nature Reviews Materials · 引用 2,398 · [DOI](https://doi.org/10.1038/s41578-019-0165-5)

79. 面向固态电池量产的有机—无机复合材料调控

Tailoring inorganic – polymer composites for the mass production of solid-state batteries

2021 · Nature Reviews Materials · 引用 1,042 · [DOI](https://doi.org/10.1038/s41578-021-00320-0)

80. 为先进电池化学设计聚合物

Designing polymers for advanced battery chemistries

2019 · Nature Reviews Materials · 引用 911 · [DOI](https://doi.org/10.1038/s41578-019-0103-6)

10. 分析化学 / 传感检测

81. 基于便携式拉曼光谱仪的表面增强拉曼光谱应用：近期进展综述

Applications of surface - enhanced Raman spectroscopy based on portable Raman spectrometers: A review of recent developments

2022 · Luminescence · 引用 48 · [DOI](https://doi.org/10.1002/bio.4383)

82. 石墨烯基表面增强拉曼散射（SERS）传感：基于文献计量的分析与综述

Graphene-Based Surface-Enhanced Raman Scattering (SERS) Sensing: Bibliometrics Based Analysis and Review

2022 · Chemosensors · 引用 23 · [DOI](https://doi.org/10.3390/chemosensors10080317)

83. 多功能纳米试剂与基于SERS的多模态传感用于食品生物毒素检测的进展

Advances in Multifunctional Nanoagents and SERS-Based Multimodal Sensing for Biotxin in Foods

2025 · Foods · 引用 17 · [DOI](https://doi.org/10.3390/foods14081393)

84. 表面增强拉曼散射光纤传感器：进展与展望

Surface - Enhanced Raman Scattering Optical Fiber Sensors: Advances and Prospects

2025 · Journal of Raman Spectroscopy · 引用 4 · [DOI](https://doi.org/10.1002/jrs.70058)

85. 表面增强拉曼光谱

Surface-enhanced Raman spectroscopy

2022 · Nature Reviews Methods Primers · 引用 894 · [DOI](https://doi.org/10.1038/s43586-021-00083-6)

86. 比率型光学纳米探针实现精确的分子检测与成像

Ratiometric optical nanoprobes enable accurate molecular detection and imaging

2018 · Chemical Society Reviews · 引用 822 · [DOI](https://doi.org/10.1039/c7cs00612h)

87. 超越热点工程的表面增强拉曼散射（SERS）平台设计：分析物操控与杂化材料中的新兴机遇

Designing surface-enhanced Raman scattering (SERS) platforms beyond hotspot engineering: emerging opportunities in analyte manipulations and hybrid materials

2018 · Chemical Society Reviews · 引用 764 · [DOI](https://doi.org/10.1039/c7cs00786h)

11. MOF / 多孔材料

88. ZIF-8基MOF衍生物用于电催化的近期进展

Recent progress on ZIF-8 based MOF derivatives for electrocatalysis

2024 · Coordination Chemistry Reviews · 引用 142 · [DOI](https://doi.org/10.1016/j.ccr.2023.215492)

89. 金属有机框架—纤维素复合材料的设计策略与应用综述

Review on design strategies and applications of metal-organic framework-cellulose composites

2022 · Carbohydrate Polymers · 引用 109 · [DOI](https://doi.org/10.1016/j.carbpol.2022.119539)

90. 宏观尺寸金属有机框架构筑体的制备：综述

Fabrication of metal-organic framework architectures with macroscopic size: A review

2022 · Coordination Chemistry Reviews · 引用 78 · [DOI](https://doi.org/10.1016/j.ccr.2022.214520)

91. 金属—空气电池氧电极用金属有机框架衍生双功能电催化剂的发展综述

A review on development of metal – organic framework-derived bifunctional electrocatalysts for oxygen electrodes in metal – air batteries

2023 · RSC Advances · 引用 66 · [DOI](https://doi.org/10.1039/d2ra06741b)

92. 钴酞菁在电催化CO₂还原制CO中的进展：简要综述

Advances of Cobalt Phthalocyanine in Electrocatalytic CO₂ Reduction to CO: a Mini Review

2022 · Electrocatalysis · 引用 41 · [DOI](https://doi.org/10.1007/s12678-022-00766-y)

93. 金属有机框架用于水电解的近期进展：综述

Recent Advancement in Metal – Organic Framework for Water Electrolysis: A Review

2022 · ChemNanoMat · 引用 27 · [DOI](https://doi.org/10.1002/cnma.202200115)

94. 共价有机框架：设计、合成与功能

Covalent Organic Frameworks: Design, Synthesis, and Functions

2020 · Chemical Reviews · 引用 4,083 · [DOI](https://doi.org/10.1021/acs.chemrev.9b00550)

95. 金属有机框架（MOF）基与MOF衍生纳米催化的现状与展望

State of the Art and Prospects in Metal – Organic Framework (MOF)-Based and MOF-Derived Nanocatalysis

2019 · Chemical Reviews · 引用 2,451 · [DOI](https://doi.org/10.1021/acs.chemrev.9b00223)

96. 导电金属有机框架

Electrically Conductive Metal – Organic Frameworks

2020 · Chemical Reviews · 引用 2,024 · [DOI](https://doi.org/10.1021/acs.chemrev.9b00766)

12. 计算化学 / 理论模拟

97. 氢键二元液晶混合物的电子与构象分析（HOMO-LUMO、MEP、NBO、ELF、LOL、AIM）：DFT/TD-DFT方法

Electronic and structure conformational analysis (HOMO-LUMO, MEP, NBO, ELF, LOL, AIM) of hydrogen bond binary liquid crystal mixture: DFT/TD-DFT approach

2022 · Computational and Theoretical Chemistry · 引用 83 ·
[DOI](https://doi.org/10.1016/j.comptc.2022.113920)

98. 借Bader电荷与态密度分析对氧空位增强NiO反应活性的第一性原理研究

First-principles investigations into the effect of oxygen vacancies on the enhanced reactivity of NiO via Bader charge and density of states analysis

2023 · Catalysis Today · 引用 81 · [DOI](https://doi.org/10.1016/j.cattod.2022.01.012)

99. 纳米材料表面/界面高精度机器学习原子间势的构建——综述

Construction of High Accuracy Machine Learning Interatomic Potential for Surface/Interface of Nanomaterials—A Review

2023 · Advanced Materials · 引用 79 · [DOI](https://doi.org/10.1002/adma.202305758)

100. 水电解电催化的互补计算线索：DFT与机器学习视角

Complimentary Computational Cues for Water Electrocatalysis: A DFT and ML Perspective

2023 · Advanced Functional Materials · 引用 72 · [DOI](https://doi.org/10.1002/adfm.202312425)

101. 计算电催化中的气相误差：综述

Gas-phase errors in computational electrocatalysis: a review

2024 · EES Catalysis · 引用 71 · [DOI](https://doi.org/10.1039/d3ey00126a)

102. 从密度泛函理论到概念密度泛函理论与生物体系

From Density Functional Theory to Conceptual Density Functional Theory and Biosystems

2022 · Pharmaceuticals · 引用 55 · [DOI](https://doi.org/10.3390/ph15091112)

103. 材料与分子的高斯过程回归

Gaussian Process Regression for Materials and Molecules

2021 · Chemical Reviews · 引用 1,268 · [DOI](https://doi.org/10.1021/acs.chemrev.1c00022)

104. 合金的机器学习

Machine learning for alloys

2021 · Nature Reviews Materials · 引用 719 · [DOI](https://doi.org/10.1038/s41578-021-00340-w)

105. 轻质烷烃C-H键活化：理论视角

C-H bond activation in light alkanes: a theoretical perspective

2021 · Chemical Society Reviews · 引用 374 · [DOI](https://doi.org/10.1039/d0cs01262a)

使用建议

第一步：从12个方向里挑3个感兴趣的，各读1篇的摘要+展望。

第二步：留下「还想再读」的1-2个，找本院对应老师聊。

第三步：老师会告诉你该读什么——比任何清单都准。

编制于 2026-09-22。译文仅供参考，检索以英文原题为凭。